OOAD Mini Project

Food Delivery App

Team Details : TEAM 12 - OVERRIDERS
Shashank S Shetty - PES1UG23AM284
Shashank H S - PES1UG23AM282
Shivakumar S Kokatanur - PES1UG23AM286
Sachith Kanwa - PES1UG23AM253

Dispatch Assignment System – Project Synopsis

1. Executive Summary

This document outlines the design and implementation strategy for the **Dispatch Assignment module**. The primary goal of this module is to assign a delivery partner to an order based on availability while ensuring clear failure handling when no partner is available.

The system enforces **strict dispatch rules**, guarantees **only available partners are selected**, and provides **explicit reasons for assignment failure**. The module is designed to integrate seamlessly with order and delivery partner services.

2. Requirements to Satisfy

To ensure proper functionality and adherence to system architecture, the following requirements must be met:

- **Availability Constraint:** Only delivery partners marked as *available* can be assigned.
- **Failure Handling:** If no partner is available, dispatch must fail with a clear reason.
- **Deterministic Assignment:** The assignment rule must be consistent and testable.
- **Integration:** Must work with Order and Delivery modules.
- **Compliance:** Must pass all unit tests and expose clear service contracts.

3. Proposed Implementation

The implementation follows a **clean architecture pattern** within the package:

edu.classproject.dispatch

3.1 Components

- **DispatchService (Interface):**
  Defines the contract for assigning delivery partners.
- **DefaultDispatchService (Implementation):**
  Contains the core business logic for partner assignment.
- **DispatchAssignment (Record):**
  Represents the result of the dispatch process.
- **Dependencies:**
  - DeliveryPartnerService → to fetch available partners
  - OrderService → to validate order existence

3.2 Contract Types (DTOs)

- **DispatchAssignment**

assignmentId
orderId
partnerId
reason

3.3 Utilities

- **PartnerSelector:**
  Handles selection logic (e.g., first available / nearest partner).

4. Functional Behavior

Feature	Description
assignPartner	Assigns an available partner to an order
success case	Returns assignment with partnerId
failure case	Returns null partnerId with failure reason

5. Validation Rules

Field	Validation Requirement
Order ID	Must be non-null and correspond to an existing order in the system.
Restaurant ID	Must be non-null and valid for the given order.
Customer ID	Must be non-null and associated with a valid user.
Available Partners List	Must be fetched from DeliveryPartnerService and should not be null.
Partner Availability	Only partners with available = true can be considered for assignment.
Partner Selection	At least one available partner must exist to proceed with assignment.
Assignment ID	Must be generated uniquely for every successful dispatch.
Failure Condition	If no partner is available, the system must return a failure with reason: *"No delivery partner available"*.
Consistency Rule	Given the same set of available partners, the selection logic must produce consistent results.

6. Test and Demo Plan

6.1 Unit Tests

- **Assignment Success:**
  Verify partner is assigned when available.
- **Assignment Failure:**
  Verify failure when no partners are available.
- **Correct Partner Selection:**
  Ensure only available partner is chosen.
- **Edge Case:**
  Empty partner list should return failure.

6.2 Demo Scenario

1. Create an order.
2. Add delivery partners:
   - P1 (Available)
   - P2 (Busy)
3. Call dispatch service.
4. System assigns P1.
5. If all partners are busy → failure message returned.

7. Proposed Class Diagram